

COURSE TITLE : GREEN PRINTING AND WASTE MANAGEMENT
COURSE CODE : 6104
COURSE CATEGORY : E
PERIODS/ WEEK : 5
PERIODS/ SEMESTER : 75
CREDIT : 5

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Green printing	20
2	Material and time waste reduction	20
3	Advanced press control	18
4	waste management	17
TOTAL		75

Course Outcomes:

G .O.	ON THE COMPLETION OF THE STUDY OF THIS MODULE STUDENT WILL BE ABLE :
1	To understand Green printing
2	To comprehend waste reduction
3	To comprehend advanced presses
4	To understand waste management in printing industry

MODULE I GREEN PRINTING

1.1.0 To understand Green printing

- 1.1.1. To explain green printing
- 1.1.2. To explain eco friendly paper
- 1.1.3. To explain eco friendly ink
- 1.1.4. To explain eco friendly offset plates
- 1.1.5. To explain eco friendly chemicals

MODULE II MATERIAL AND TIME WASTE REDUCTION

2.1.0 To comprehend waste reduction

- 2.1.1 To explain printing industry waste
- 2.1.2 To describe waste reduction during print production
- 2.1.3 To describe make ready time reduction methods

MODULE III ADVANCED PRESS CONTROL

3.1.0 To comprehend advanced presses

- 3.1.1 To explain advanced press control consol

- 3.1.2 To explain press inline measurement systems
- 3.1.3 To explain press automation

MODULE IV WASTE MANAGEMENT

4.1.0 To understand waste management in printing industry

- 4.1.1 To explain types of wastes
- 4.1.2 To explain recycling of materials and consumables
- 4.1.3 To explain waste management

CONTENT DETAILS

MODULE I

Definition of green printing. Eco friendly printing paper - types– recycled paper, wood free paper, etc.
Eco friendly inks – types - vegetable based ink, bio inks etc. Ink recycling
Eco friendly offset plates – low chemical , chemical free, etc. recycling of plates.
Eco friendly chemicals – IPA substitutes, bio chemicals etc.
Eco friendly production - waste reduction, waste management.

MODULE II

Types of waste produced in – screen printing, offset printing, flexographic printing, gravure printing, digital printing.

Ways to reduce waste during production in – screen printing, offset printing, flexographic printing, gravure printing, and digital printing.

Optimization. Make ready time. Make ready time comparison between manual and automated systems. Ways to reduce make ready time in sheet fed and web fed machines.

MODULE III

Advanced presses - press remote control consol, Ink feed preset systems, color measurement and control system, Register measurement and control, inline print quality measurement and control systems, The State of the Art in Remote Control, Measurement and Control Systems.

Automation in print media production – wash up procedure, plate changing, sheet size and paper travel adjustments, Ink Feed Presetting in the Printing Unit, Ink and Dampening Solution Supply, Make ready for Print Production.

MODULE IV

Types of waste – solid, water, air. Reduce – Reuse – Recycle.

Recycling of print materials and consumables.

Printing waste management. Solid waste management. Solvent management. Ink management. Hazardous waste management. Air emission management. Noise management. Water and energy consumption.

Reference :		
Author	Title	Publishers
Michael Bernard, John Peacock.	Handbook of printing and production	
Helmut Kipphan	Hand book of print media	Springer Science & Business Media
Kevin kooper	Lean printing Path way to success	PIA GATF Press
HL Apfelberg and MJ Apfelberg	Implementing quality management in graphic arts	GATF

